Riding the Wave?

Who Starts, Follows and Gains from ML Research Trends

A Pre-registered Study of ICLR Submissions, 2017–2026:
43,480 Discovery and 11,099 Held-out Papers

[Your name]

[Department], [Institution]

[Venue / seminar] [Date]

Outline

1. Motivation and Research Questions
2. Data and Measurement
3. Methodological Contribution: Hindsight Bias in Topic Maps
4. Empirical Strategy
5. Results
6. Held-out Replication
7. Conclusion

Motivation: Research Fashions in Machine Learning

Share of ICLR submissions that ...	2017	2026
propose a new architecture	24%	8%
build on a foundation model	0.3%	54%
carry a branded title ("NAME: subtitle")	9%	41%

Table 1: Field-wide shifts in content and in style.

- ML research moves in waves, in *what* is studied and in *how* it is presented.
- Are these fashions? If so: what do they displace, who drives them, and are followers rewarded?

Research Questions and Pre-registered Hypotheses

Question	Key hypothesis	Predicted sign
RQ1 Rise: what does a fashion displace?	**H1d**: prior foundation-model (FM) adoption in a topic → share of new-architecture papers	−
RQ2 Following: who introduces new work?	**H2b**: newcomer paper → new-method paper	−
	H2c: newcomer paper → benchmark/evaluation paper	+
RQ3 Reward: are followers rewarded?	**H3a**: topic maturity (prior papers in topic) → acceptance	−

Table 2: Selected hypotheses. All ten registered hypotheses and their verdicts: Appendix A1.

Our RQ2 prediction followed the fashion view: newcomers imitate, insiders innovate.

Related Work and Positioning

Two views of who drives new work

- *Fashion and herding*: ideas spread by imitation (Abrahamson, 1991; Bikhchandani et al., 1992) $\Rightarrow$ newcomers follow
- *Outsider innovation*: new entrants bring new ideas (Azoulay et al., 2019); tradition vs. innovation (Foster et al., 2015)
- Team size: large teams develop, small teams disrupt (Wu et al., 2019)

Meta-research on ML

- Benchmark concentration (Koch et al., 2021)
- Values encoded in ML papers (Birhane et al., 2022)

Measurement challenges we address

1. Survivorship: only published papers observed
2. Hindsight: topics defined on the full period
3. Forking paths (Gelman & Loken, 2014) $\Rightarrow$ pre-registration (Nosek et al., 2018)

Data and Measurement

Construct	Operationalisation	Validation
Corpus	43,480 discovery + 11,099 held-out ICLR submissions (released former confirmation split), 2017–2026, from OpenReview: accepted, rejected, withdrawn	Held-out: never analysed before its single registered run; a pre-registered out-of-sample replication, not a certified untouched holdout
Topic	SPECTER2 embeddings of title + abstract (Singh et al., 2023); $k = 100$ clusters; held-out papers placed on the frozen maps	Agreement with ICLR areas: NMI 0.30 (keyword baseline 0.27); robust to $k = 75, 150$
Contribution type	LLM with a fixed codebook: method, benchmark/evaluation, empirical, theory, other	Blind second pass: $\kappa = 0.93$ on both splits (held-out: 555 papers)
Newcomer paper	No author has an earlier ICLR submission *in the topic* (73% of papers)	Exact OpenReview author IDs
Acceptance	Final decision	Sealed in code until each split's single registered run (held-out: the earlier keyword grid's overlap is unknown)

Table 3: Constructs, measures and checks.

Hindsight Bias in Topic Maps

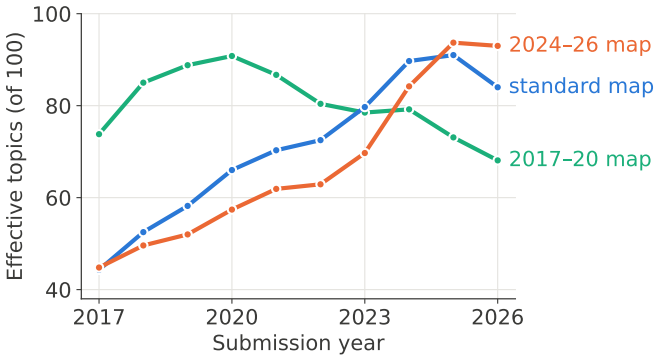


Figure 1: Effective number of topics per year under three $k = 100$ maps fitted on different years.

- 69% of submissions are from 2024–26: a map fitted on all years is a map of the *present*.
- Each map finds peak diversity in the years used to fit it.

Implication

Topic maps are not neutral preprocessing: the fitting window shapes the history one observes.

Consequence: Substantive Conclusions Depend on the Map

- Emerging topics ($\geq 3\times$ growth): **8** with the 2017–20 map vs. **29** with the standard map.
- Small topics 'predict' 2-year growth on the standard map ($R^2 = 0.28$), not on the 2017–20 map (-0.03). (Post hoc.)
- Do topics age, from new methods towards benchmarks? Same papers, same model:

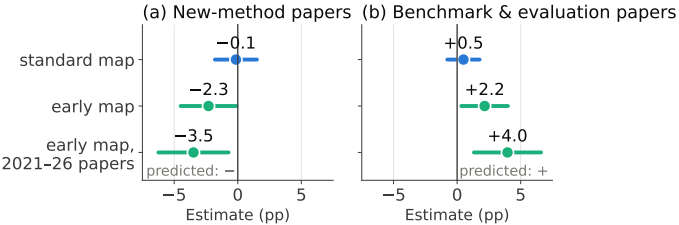


Figure 2: Coefficient on topic maturity (log prior papers), 95% CI. Early-map rows are exploratory.

⇒ Every registered result must also pass two hindsight checks (early-year and year-balanced maps).

Empirical Strategy

Specification (linear probability model)

$$y_i \;=\; \beta\, x_i \;+\; \mu_{k(i)} \;+\; \tau_{t(i)} \;+\; \varepsilon_i$$

- $y_i \in \{0,1\}$: paper type or acceptance; x_i: e.g. newcomer paper
- μ_k, τ_t: topic and year fixed effects (topic $\times$ year for RQ2)
- Standard errors clustered by topic (100 clusters)

Decision rule (registered). Supported only if:

1. predicted sign, and Holm-adjusted $p < 0.05$;
2. same sign in $\geq 75\%$ of 9–11 alternative specifications;
3. same sign under both hindsight checks.

Protocol

Pre-registered; variables built blind to acceptance; 12,400 simulation runs; code and inputs frozen (SHA-256); a single run; every estimate reported.

Simulation finding

No-effect data: clustered SEs rejected up to 12.6% at nominal 5%, so RQ2 gets two extra checks (two-way clustering, a stress test).

Held-out rule (registered)

Registered tests only; one-sided $\alpha = 0.05$; frozen code and topic maps; features built on discovery + held-out papers.

RQ1: Foundation-Model Adoption and New Architectures

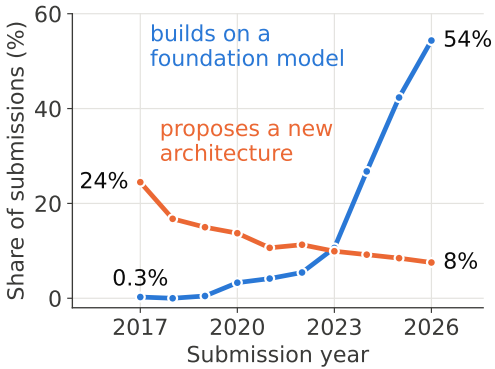


Figure 3: Field-wide trends (descriptive).

Within topic, net of year effects: prior FM adoption vs. new-architecture papers.

	Discovery	Held-out
$\hat{\beta}$ (0 $\rightarrow$ 100% FM)	-0.080	$\mathbf{-0.110}$
(SE)	(0.021)	(0.036)
95% CI	$[-0.121, -0.038]$	$[-0.181, -0.039]$
p	Holm 0.001	one-sided 0.001
Same sign (specs)	9 / 9	9 / 9
N	42,990	10,978

- H1d **supported**, and **replicated** on the held-out papers.
- Topics at 50% vs. 10% FM adoption: about one third fewer new-architecture papers (discovery; base 9%).

RQ2: Newcomers and Contribution Types

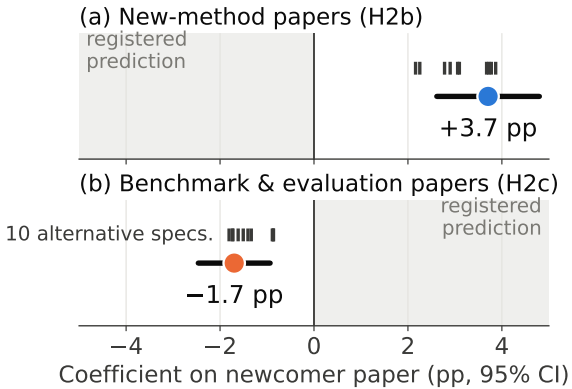


Figure 4: Discovery; topic × year FE; grey: predicted side.

	H2b (method)	H2c (bench.)
Discovery $\hat{\beta}$ (SE)	$+0.037$ (0.006)	-0.017 (0.004)
Same sign	10/10	10/10
Held-out $\hat{\beta}$	$\mathbf{+0.036}$	$\mathbf{-0.021}$
95% CI	$[0.015, 0.057]$	$[-0.033, -0.008]$
Base rate: disc.	75%	7.7%
held-out	75%	7.9%

- Discovery: both opposite to prediction (wild cluster bootstrap $p < 0.001$; $N = 42{,}315$)
- Held-out: reversed predictions H2b′/H2c′ (registered) **supported** (one-sided Holm $p = 0.001$ each; $N = 10{,}718$)
- Consistent with outsider innovation, not with herding

Most of the H2c gap goes with team size; H2b does not.

RQ3: Is Following the Fashion Rewarded?

Association with acceptance	Discovery	Held-out	Status
Topic maturity, per log unit of prior papers (H3a)	-1.7 pp (SE 1.0)	-1.5 pp (SE 1.9)	inconclusive in both
Branded title	ratio 1.02	not tested	exploratory
New arch. $\times$ prior FM adoption (0 $\rightarrow$ 100%)	$+8.3$ pp (SE 3.5)	not tested	exploratory, $p = 0.02$

Table 4: Topic and year fixed effects; SEs clustered by topic; decided papers, 2018–2026 (H3a: $N = 32{,}190$, MDE_{80} 4 pp; held-out $N = 8{,}185$, MDE_{80} 4.8 pp, one-sided $p = 0.21$).

Reading

No detectable reward for following the fashion at review; the crowding test is inconclusive in both splits. If anything, papers that go against the tide (architectures where FMs dominate) do relatively better (exploratory).

Other correlates of acceptance (theory papers, team size): Appendix A7–A8.

Far Jumps Go with Lower Acceptance (Post Hoc, Replicated)

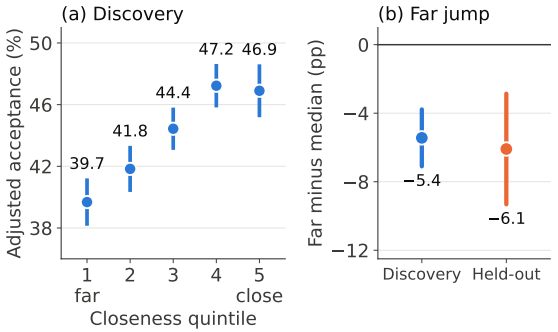


Figure 5: Decided papers with ≥ 1 experienced author, 2023–26 (E1 sample: discovery 20,430; held-out 5,344). Closeness: highest cosine between the abstract and an experienced co-author's earlier ICLR work. (b) 10th percentile minus median. 95% CIs.

- Far jump from the team's own record vs. median: -5.4 pp in discovery; -6.1 pp held-out (CI -9.3 to -2.9; Holm $p = 0.005$)
- Same author, closer paper: $+0.55$ pp (discovery), $+1.06$ pp (held-out) per $+0.1$ cosine
- Near vs. median: not lower ($+1.7$ pp discovery; $+2.3$ pp held-out, not registered)
- **Not replicated**: the field-novelty floor. Least-novel fifth: -3.3 pp in discovery, -0.1 pp held-out (CI -3.5 to $+3.2$); uninformative by the registered rule (power 0.47, MDE_{80} 5.4 pp)

Post hoc in discovery; held-out tests registered before the run. Held-out: post-decision abstracts, no cleaned-text check. Associations: co-author choice and failed projects not controlled.

Why Do Some Areas Lose Favour? (Post Hoc)

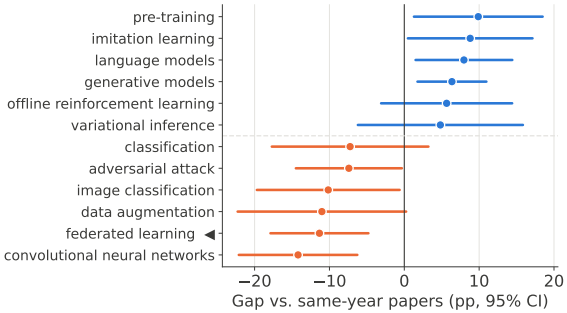


Figure 6: Acceptance gap vs. papers of the same year, 2018–2026: six highest and six lowest author keywords (≥ 5 years of data; 68 keywords, 32,190 decided papers).

Pattern: areas differ by about ± 10 pp; generative and pre-training work above, maturing areas (CNNs, image classification, federated learning) below.

Three explanations, checked on the clearest case (federated learning, -11 pp):

- *Weaker authors moving in?* No: authors became more established (4% $\rightarrow$ 60%); controls move the gap little.
- *The field moving on?* Yes: share peaked in 2023 (3.2%), 1.1% by 2026.
- *Higher bar for a cooling area?* Consistent: gap of -11 to -15 pp every year since 2023.

No single area survives correction for 68 comparisons. Held-out: area edges shrink to $0.60\times$ but persist (topics); the FL gap points the same way (-4.9 pp) but is not confirmed (A19).

Do the Results Hold on Held-out Papers?

Test	Pred.	Discovery	Held-out [95% CI]	One-sided p	Verdict
Registered hypotheses (one-sided p, Holm-adjusted within family)					
H1d FM adoption ($0 \to 1$) $\to$ new arch.	$-$	-8.0	-11.0 [-18.1, -3.9]	0.001	supported
H2b′ newcomer $\to$ method paper	$+$	$+3.7$	$+3.6$ [$+1.5$, $+5.7$]	0.001	supported
H2c′ newcomer $\to$ benchmark paper	$-$	-1.7	-2.1 [-3.3, -0.8]	0.001	supported
H3a maturity (log) $\to$ acceptance	$-$	-1.7	-1.5 [-5.2, $+2.2$]	0.21	inconclusive†
H4-FL FL keyword $\to$ acceptance	$-$	-11.3	-4.9 [-15.4, $+5.7$]	0.14	not supported†
PH5 primary (post hoc in discovery; one-sided p, Holm-adjusted over the pair)					
E1 far jump (p10 $-$ median)	$-$	-5.4	-6.1 [-9.3, -2.9]	0.005	supported
P1b least-novel $-$ middle fifth	$-$	-3.3	-0.1 [-3.5, $+3.2$]	0.47	not supported†
PH5 secondary (one-sided p, unadjusted)					
E4 within author, per $+0.1$ cos	$+$	$+0.55$	$+1.06$ [$+0.23$, $+1.89$]	0.006	supported
Teams H1 one insider vs none	$+$	$+3.4$	$+3.9$ [$+0.3$, $+7.5$]	0.017	supported
A1 Emerging $-$ Mainstream keyword	$+$	$+2.7$	$+3.3$ [-0.2, $+6.7$]	0.031	supported
Premia b topic edges (early map)	$0 < b < 1$	0.38‡	0.60 [0.32, 0.88]	0.003	supported

Table 5: 11,099 released former confirmation papers this study had never analysed: a pre-registered out-of-sample replication, not a certified untouched holdout. Supported: one-sided $p < 0.05$, Holm within families (so a 95% CI can cross 0: A1). pp unless noted. †Below MDE_{80}: uninformative. ‡Within-discovery slope (A17). Also registered (A14): H4-FL trend, P7 not supported; premia EB supported.

Contributions

1. **Methodological.** Topic maps fitted with hindsight manufacture emerging topics and can reverse trend conclusions; early-year maps provide a cheap check.
2. **Design.** A pre-registered, outcome-blind, simulation-checked, single-run protocol for science-of-science studies, with a registered held-out replication.
3. **Substantive** (associations).
   - **Replicated:** where foundation models spread, new architectures fade (H1d); newcomer papers lean towards new methods and away from benchmarks (H2b′/H2c′); far jumps from the team's own record go with lower acceptance (post hoc in discovery).
   - **Not replicated:** the field-novelty floor (least-novel fifth); the held-out test was uninformative (power 0.47, MDE_{80} 5.4 pp).
   - **Not confirmed:** the federated-learning acceptance gap; the held-out test was uninformative (134 papers, MDE_{80} 14.5 pp).
   - **Inconclusive** in both splits: crowded (mature) topics and acceptance.

Limitations and Future Work

Limitations

- One venue (ICLR)
- Associations, not causal effects
- LLM labels: consistent ($\kappa = 0.93$), not yet human-validated
- Camera-ready text: the discovery cleaned-text checks held; no such check for held-out papers
- Held-out papers are a released former holdout, not certified untouched
- Pooled re-run (54,579 papers) is exploratory and confirms nothing (A18)
- 2021 author-ID change undercounts incumbents (expected bias towards zero)

Future work

- A prospective test on future ICLR submissions
- Extension to ACL Anthology / ARR
- Human validation of contribution labels
- Review scores: reviewers vs. final decisions

Thank you

Questions and comments welcome

[email]

A1. All Pre-registered Hypotheses

	Prediction	Estimate (95% CI)	Verdict
H1a	fewer new methods in mature topics	-0.1 pp $(-1.8, +1.5)$	not supported
H1b	more benchmarks in mature topics	$+0.5$ pp $(-0.8, +1.8)$	not supported
H1c	more empirical studies in mature topics	-0.3 pp $(-1.3, +0.7)$	not supported
H1d	**fewer new architectures after FM adoption**	$\mathbf{-8.0}$ pp $(-12.1, -3.8)$	**supported**
H1e	more empirical/evaluation after FM adoption	-0.6 pp $(-3.6, +2.4)$	not supported
H2a	fewer incumbents in hot topics	-0.7 pp $(-2.7, +1.2)$	inconclusive
H2b	**newcomers: fewer new methods**	$\mathbf{+3.7}$ pp $(+2.6, +4.8)$	**significant, opposite**
H2c	**newcomers: more benchmarks**	$\mathbf{-1.7}$ pp $(-2.5, -0.9)$	**significant, opposite**
H3a	lower acceptance in mature topics	-1.7 pp $(-3.7, +0.3)$	inconclusive (low power)
H3b	… more so for method papers	-0.4 pp $(-1.5, +0.8)$	inconclusive

Discovery record. Maturity effects are per log unit of prior papers in the topic (2.7× more). H2d/H3c (buzzwords) were dropped before any estimate because the measure failed its blind validation. Held-out tests of H1d, H2b′, H2c′ (the reversed predictions), H3a and H4-FL: slide 16 and A14.

A2. RQ2 Robustness

	New methods	Benchmarks & eval.
Main estimate	$+3.7$ pp	-1.7 pp
Range over 10 alternative specifications (all $p < 0.05$)	$+2.2$ to $+3.9$	-1.8 to -0.9
Two-way clustering (topic $\times$ early-map topic)	$+3.7$	-1.7
Team size controlled	$+3.9$	-0.5 ($p = 0.07$)
Only years with consistent author IDs	$+3.7$	-1.7
Excluding the paper's own earlier versions	$+3.8$	-1.7
Corrected for label errors	$+4.1$	-1.8

Base rates: 75% of papers are new methods, 7.7% benchmark or evaluation; 73% are newcomer papers. Confirmatory p from a wild cluster restricted bootstrap (Cameron et al., 2008): $p < 0.001$ for both. The method gap appears in every era (2019–20: $+5.5$ pp).

A3. Calibration of Standard Errors

- Before seeing any result: 12,400 simulation runs, including no-effect scenarios with realistic topic structure.
- When true topic structure cuts across the 100 topics, the standard test rejected 12.6% (H2b) and 10.0% (H2c) of the time at nominal 5%. The wild bootstrap did not fix this.
- Added before the run:
  - two-way clustering (two topic maps at once);
  - a stress test that deflates each t-statistic by the simulated inflation (1.31 / 1.22).
- RQ2 results: $t = 6.7$ and -4.4. After the stress test, Holm $p = 4 \times 10^{-6}$ and 1×10^{-3}.

Lesson

Check the test on data shaped like yours before trusting its p-values.

A4. Validity of Topics and Labels

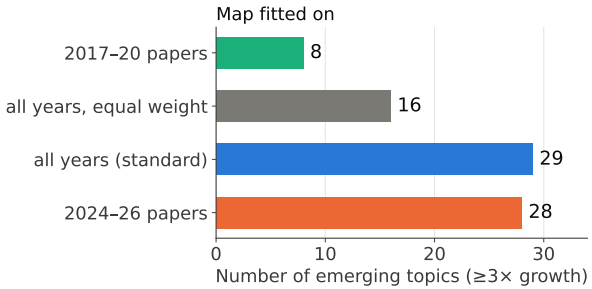
Topics (SPECTER2, 100 clusters)

- Agreement with ICLR's primary areas (2024–26): NMI 0.30 vs. 0.27 for a TF-IDF baseline
- Blind naming test: from title and abstract, a checker picks the right topic 80% of the time at the 100-topic level (61% among 198 fine topics; chance 10%)
- Stable across refits (ARI 0.66)

Contribution labels (LLM, fixed codebook)

- Blind second pass on 2,174 papers: $\kappa = 0.93$
- Held-out papers (same model and prompt): second pass on 555 papers, $\kappa = 0.93$
- 95.8% agreement with 2,300 reference labels (also LLM-made, so this shows consistency, not human validity)
- A correction for label errors makes the RQ2 estimates slightly larger

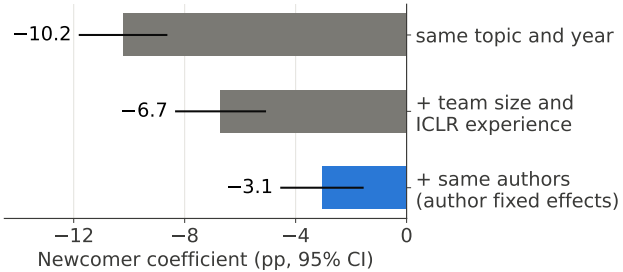
A5. Hindsight: Further Evidence



- Share of 2024–26 papers in emerging topics: 22% with the early map vs. 41% with the standard map
- Declining topics: 13 with the standard map vs. 26 with the year-balanced map, so faded topics are absorbed into today's topics
- Fit of papers to their topic: standard map 0.62 (2017) vs. 0.65 (2026); early map 0.69 vs. 0.53

All maps have 100 topics; only the fitting years differ.

A6. Newcomer Acceptance Gap (Exploratory)

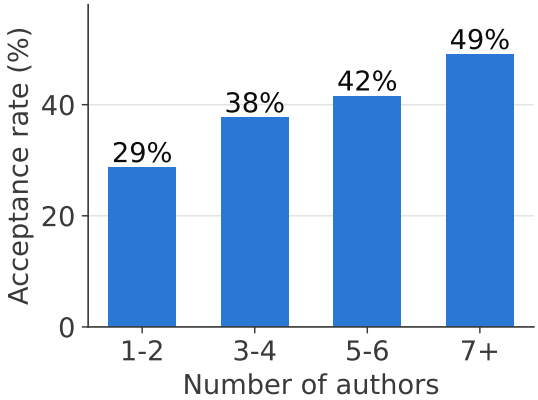


- Raw acceptance: 37% vs. 47%
- The gap shrinks as comparisons tighten; within the same authors it is about -3 pp
- The same person entering a topic new to them: -1.4 pp

Mostly about *who* writes newcomer papers. ICLR is double-blind, so the remainder need not be bias: being new to a topic can make a paper weaker.

Author-FE sample: -5.4 pp without, -3.1 pp with author fixed effects.

A7. Team Size and Acceptance (Exploratory)



- $+8.8$ pp per doubling of team size, within topic $\times$ year
- Steepest for benchmark and evaluation papers ($+13.5$ pp each); this ranking is only suggestive
- Flattest for theory (1–2 authors: 42%)
- Positive in all 20 areas; flattest in federated learning and time series

Team size also proxies compute, data and seniority (Wu et al., 2019). Who is on the team: A15.

A8. Acceptance by Contribution Type (Exploratory)

Within topic × year, relative to new-method papers:

Theory	$+8.9$ pp
Dataset / benchmark	$+3.9$ pp
Evaluation	-1.9 pp (n.s.)
Empirical study	-2.7 pp
Other (systems, applications, position papers, . . .)	-8.0 pp

Papers building on a foundation model: $+4.6$ pp vs. papers that do not.

Unadjusted, exploratory. Selection is likely: who writes theory papers differs from who writes method papers.

A9. Fashion in Title Style (Exploratory)

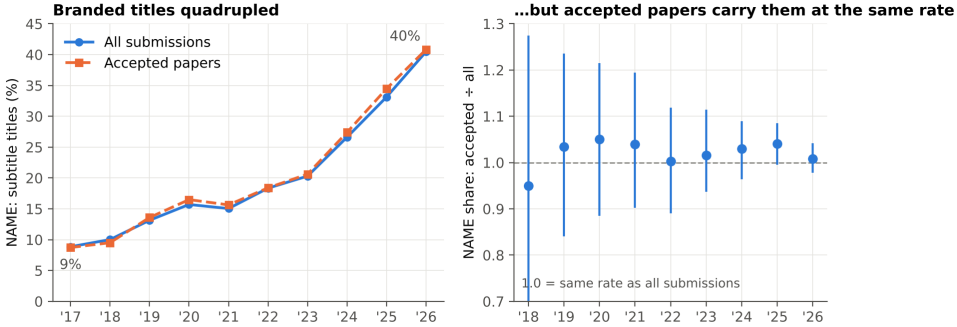
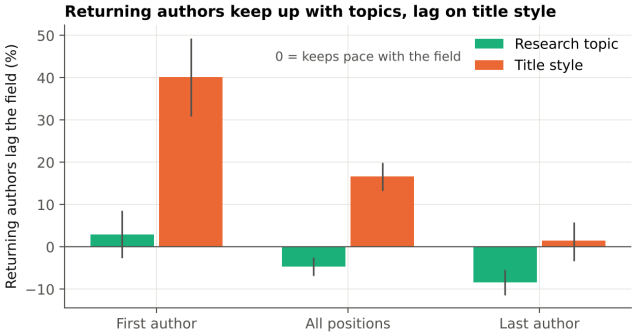


Figure 7: Left: share of branded titles. Right: share among accepted ÷ share among all submissions.

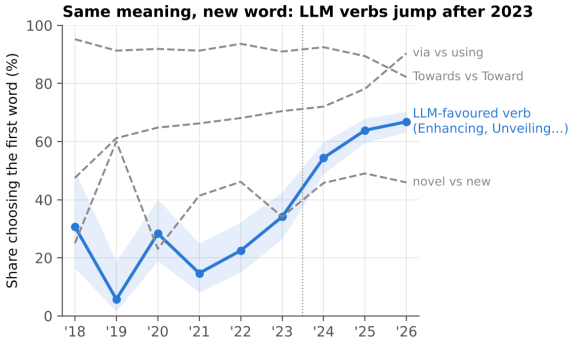
- Branded titles rose more than fourfold; accepted papers carry them at the submitted rate (ratio 1.02).
- The rise appears with and without LLM-style wording in the abstract. (Not pre-registered.)

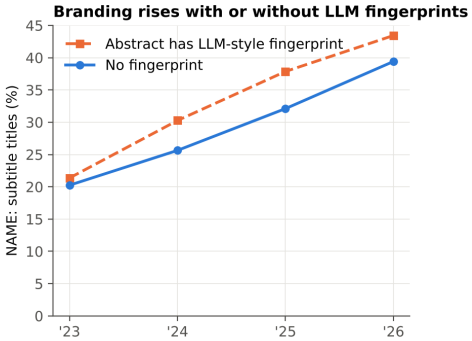
A10. Title Style: Who Carries It? (Exploratory)



- Returning authors move into new topics as fast as the field
- They lag on the new title style, most as first authors
- Topics change with the field; style changes with the people

A11. Title Style and LLM Writing Tools (Exploratory)



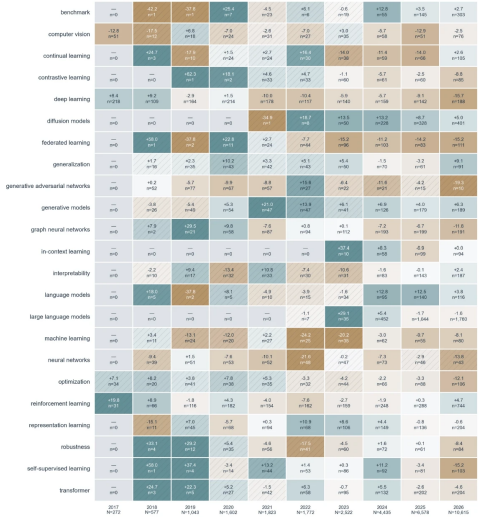


LLM-favoured words ("Enhancing", "Unveiling") jump after 2023 (odds ratio 2.2), while control word pairs do not. The "NAME: subtitle" structure rises with or without LLM fingerprints in the abstract.

A12. Definitions

Topic	one of 100 clusters of paper embeddings (title + abstract)
Topic maturity	$\log(1 +$ earlier papers in the topic$)$; $+1 = 2.7\times$ more papers
Hot topic	top fifth of topics by last year's growth
Newcomer paper	no author has an earlier ICLR submission (any outcome) in that topic
FM adoption	share of the topic's papers last year that built on a foundation model
Benchmark & eval.	primary contribution is a dataset, benchmark or evaluation
Closeness	highest cosine between the abstract and an experienced co-author's earlier ICLR submissions (their mean vector); far jump = 10th percentile
Novelty	$1-$ mean cosine to the 10 most similar earlier ICLR papers, as a percentile within topic $\times$ year
Model	linear probability model with topic and year fixed effects (topic $\times$ year cells for RQ2); errors clustered by topic
pp	percentage points

A13. Earlier Keyword × Year Grid (Exploratory, Pre-registration)



Reference for Q&A (zoom in the PDF): keyword papers minus all others, same year (pp); hatched: a group below 50 papers; 31,239 submissions.

A14. The Held-out Replication: Data, Rules and Power

- **Papers**: 11,099 with text (of 11,113 released), 2017–2026; 69% from 2024–26; 8,185 decided papers in the acceptance tests (2018–26)
- **Assignment**: the source's original confirmation split (rule not documented by the source; 1,309 pending family/link cases excluded by it); released 2026-10-18; never analysed by this study before the run
- **Status**: pre-registered out-of-sample replication; the source does not certify an untouched independent holdout. H4-FL's out-of-sample status is unverified (earlier keyword grid of unknown provenance)
- **Run**: once, hash-checked frozen code; frozen topic maps; features built on discovery + held-out papers, populations held-out only; same LLM labels (555-paper second pass, $\kappa = 0.93$)
- **Rule**: one-sided tests (p_1), $\alpha = 0.05$; Holm within families; simulated null size of the RQ tests 2.0–5.0%

Test	MDE_{80}	|Held-out est.|
H1d (0 → 100% FM)	12.0 pp	11.0 pp
H2b′	2.8 pp	3.6 pp
H2c′	2.4 pp	2.1 pp
H3a (per log unit)	4.8 pp	1.5 pp
H4-FL	14.5 pp	4.9 pp
E1	4.9 pp	6.1 pp
P1b	5.4 pp	0.1 pp
A1*	4.9 pp	3.3 pp
Teams H1*	5.1 pp	3.9 pp

*Derived post hoc, not registered. Power at the discovery size: E1 0.89 (CR1 p; slightly less under the registered rule), P1b 0.47. Reading rule: not supported with |estimate| below MDE_{80} = uninformative. Also registered: H4-FL trend $+3.2$ pp/yr (CI -3.3 to $+9.6$; $p_1 = 0.88$; opposite sign to discovery) and P7 ($+0.34$ pp per 10 novelty percentiles; $p_1 = 0.22$), not supported; premia EB ($p_1 = 0.005$), supported.

A15. Team Experience and Acceptance (Post Hoc; H1 Held-out)

Within topic × year, decided papers 2023–26:

	Discovery ($n = 24{,}601$)	Held-out ($n = 6{,}243$)
Additive model: counts of each author type; class, FM and earlier-version controls		
Per novice (no earlier ICLR submission)	$+1.5$ pp ($+1.0$, $+2.0$)	not registered
Per crossover (earlier ICLR, new to topic)	$+4.8$ pp	not registered
Per topic insider	$+5.3$ pp	not registered
Model T1: team size and number of experienced authors fixed		
One topic insider vs. none (H1)	$+3.4$ pp ($+1.2$, $+5.7$)	$+3.9$ pp ($+0.3$, $+7.5$)
Two or more insiders vs. one	$+0.5$ pp (-2.2, $+3.2$)	not registered

Discovery H1: Holm $p = 0.019$ within its family. Held-out H1 (registered, secondary): one-sided $p = 0.017$, supported. The per-head split is one specification outside the family and was not tested on held-out papers. Selection is heavy: experienced authors may join better projects, so this is not advice to recruit names. Author histories start in 2021.

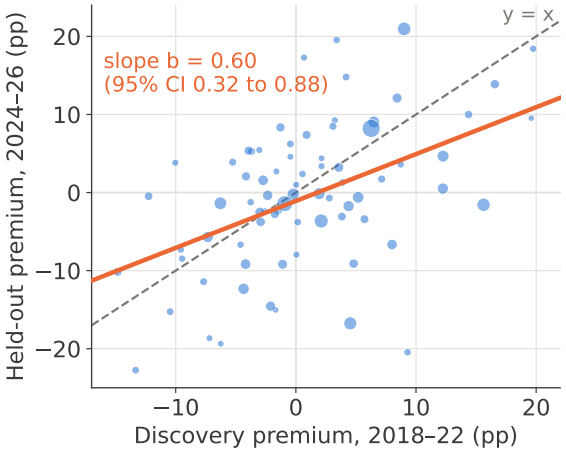
A16. Keyword Timing: Emerging vs. Mainstream (Post Hoc)

Acceptance vs. papers listing an old, flat ("Mainstream") author keyword, within topic × year:

	Discovery ($n = 22{,}302$)	Held-out
Emerging: young (1–2 years), rising (A1)	$+2.7$ pp ($+0.8$, $+4.6$)	$+3.3$ pp (-0.2, $+6.7$), supported
Cooling: old, falling (A2)	$+0.1$ pp (-2.0, $+2.3$)	not registered
Age gradient, young minus old (A3)	$+1.7$ pp, not supported	not registered
Resurgent: old, rising (nominal)	$+3.9$ pp ($+1.1$, $+6.6$)	not registered
Same keyword, rising vs. flat years	$+0.9$ pp (-0.8, $+2.7$)	not registered

Discovery A1: Holm $p = 0.020$ within its 3-test family; null-frame $p = 0.005$. Held-out A1: one-sided $p = 0.031$ (two-way clustering); null-frame $p = 0.020$. A one-sided test at 0.05 matches a 90% two-sided interval, so the 95% CI can include 0. Old rising keywords are also high, so the contrast may reflect a low Mainstream baseline as much as keyword age. Relabelling a paper is untested.

A17. Topic Edges Out of Sample (Post Hoc; Registered Test)



- 75 early-map topics, ≥ 15 decided held-out papers in 2024–26 each; dot size: papers
- Registered slope: $b = 0.60$ (CI 0.32–0.88), so $0 < b < 1$: edges persist but shrink
- Within discovery, 2018–22 $\rightarrow$ 2024–26: $b = 0.38$ (CI 0.18–0.58)
- Shrunk (EB) discovery edges predict held-out edges better than zero (one-sided $p = 0.005$)
- Not registered: keyword units $b = 0.16$ (CI -0.34 to 0.66), no detectable persistence

Premium: acceptance gap vs. same-year papers; weighted least squares. Read raw topic rankings as shrunk towards zero.

A18. Pooled Re-run (Exploratory)

Association (pp unless stated)[a]	Status	Discovery	Held-out[b]	Pooled
H1d: new architectures ∼ prior FM adoption, 0→100%	reg.	−8.0	−11.0	−8.9
H2b: method papers ∼ newcomer paper	reg., opp.[c]	+3.7	+3.6	+3.5
H2c: benchmark/eval. papers ∼ newcomer paper	reg., opp.[c]	−1.7	−2.1	−1.7
H3a: acceptance ∼ topic maturity, per log unit	reg.	−1.7	−1.5 n.s.	−1.7
E8: acceptance ∼ new arch. × prior FM, 0→100%	expl.	+8.3	—	+9.4
FL gap: keyword / topic	post hoc	−11.3 / −13.5	−4.9 n.s.† / —	−9.9† / −12.8
FL gap trend, per year: keyword / topic	post hoc	−3.0 / −2.7	+3.2 n.s.† / —	−1.9† / −1.2
Keyword extremes: CNNs / pre-training	post hoc	−14.2 / +9.9	—	−12.8 / +9.1
Far jump from own record (p10 vs p50)	post hoc	−5.4	−6.1	−5.5
Least-novel fifth vs middle fifth	post hoc	−3.3	−0.1 n.s.†	−3.0†
One topic insider vs none	post hoc	+3.4	+3.9	+2.9
Emerging vs mainstream keywords	post hoc	+2.7	+3.3	+3.1
Topic acceptance edge, 2024–26 on 2018–22: slope b	post hoc	0.38	0.60[d]	0.45

Exploratory: pooled includes the 11,099 held-out papers and is about 80% discovery papers, so agreement with discovery is expected and confirms nothing. Pooling also shifts features.

reg. = registered; expl. = planned exploratory; post hoc = PH4/PH5, post hoc in discovery (their held-out tests were registered before the run). [a] H1d–H3a, E8, far jump, least-novel fifth, insider, keyword timing: within topic and year (fixed effects). FL and keyword extremes: vs all other papers of the year. Topic edge: across topic units. [b] Registered held-out test, one-sided; n.s. = not supported; — = no held-out test. [c] opp. = opposite to the registered prediction in discovery; the held-out tests were the reversed hypotheses H2b′/H2c′. [d] Held-out 2024–26 edges on discovery 2018–22 edges: a different comparison from the other two slopes. † Registered held-out test not supported (the trend had the opposite sign); the pooled value is driven by the discovery papers.

A19. Worked Example: Federated Learning (Post Hoc)

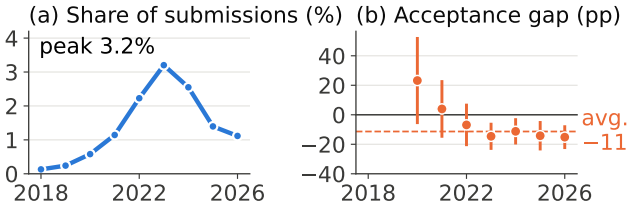


Figure 8: Federated learning (author keyword), discovery; 95% CI, years with ≥ 10 papers; dashed: average.

Held-out (registered H4-FL)

-4.9 pp (CI -15.4 to $+5.7$; one-sided $p = 0.14$): same direction, mostly from 2021–22 (2024–26: -3.6 pp); trend $+3.2$ pp/yr, not supported. **Not confirmed**: underpowered (134 FL papers; MDE_{80} 14.5 pp), so uninformative. The gap stands as discovery evidence only.

Question	Evidence (discovery)	Reading
Influx of inexperienced authors?	Authors became *more* established (4% → 60%); adjusting for type, FM and team size moves the gap little ($-11.3 \rightarrow -10.5$ pp)	Not supported
Did the field move on?	Share of submissions peaked in 2023 (3.2%), 1.1% by 2026	Share fell (outcome-free)
Lower acceptance?	Gap -11 to -15 pp each year since 2023	Gap observed; reviewers vs. papers not separable

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